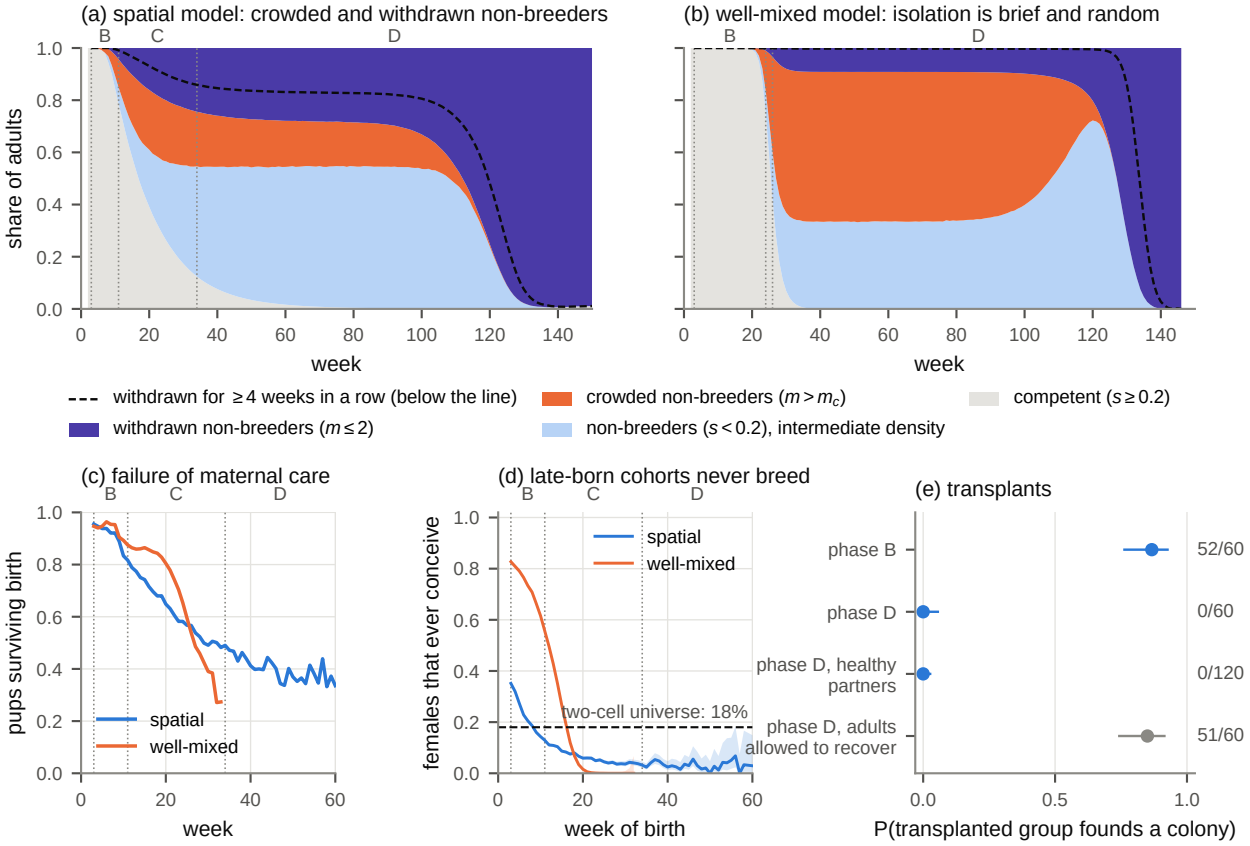
(a) spatial model: crowded and withdrawn non-breeders
B
C
D
share of adults
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
120
140
week
(b) well-mixed model: isolation is brief and random
B
D
week
withdrawn for ≥4 weeks in a row (below the line)
withdrawn non-breeders (m ≤ 2)
crowded non-breeders (m > m_c)
non-breeders (s < 0.2), intermediate density
competent (s ≥ 0.2)
(c) failure of maternal care
B
C
D
pups surviving birth
spatial
well-mixed
week
(d) late-born cohorts never breed
B
C
D
females that ever conceive
spatial
well-mixed
two-cell universe: 18%
week of birth
(e) transplants
phase B
phase D
phase D, healthy partners
phase D, adults allowed to recover
52/60
0/60
0/120
51/60
0.0
0.5
1.0
P(transplanted group founds a colony)